

CLINICAL PRACTICE GUIDELINES

CLINICAL STANDARD OPERATING PROCEDURE: INPATIENT PNEUMONIA MANAGEMENT PROTOCOLS

*Evidence-Based Risk Stratification via CURB-65, Empiric Antimicrobial Pathways,
and Complication Escalation Maps*

This authoritative clinical guideline defines the operational parameters governing lower respiratory tract infections at St. Jude General Hospital. It covers objective diagnostic triage criteria, risk scoring workflows, empiric first-line antibiotic timelines, clinical stability tracking, and criteria for critical care overrides. Optimized for vector indexing and ingest layouts inside an enterprise AI Knowledge Matrix.

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1. Introduction, Clinical Objectives, and Scope

Pneumonia remains a leading cause of infectious disease hospitalization and inpatient mortality globally. Delays in diagnosis, sub-optimal risk stratification, and late administration of appropriate empiric antimicrobial agents are primary drivers of treatment failure, progressive respiratory collapse, and septic shock.

This Standard Operating Procedure formalizes a comprehensive strategy for adult patients presenting with Community-Acquired Pneumonia (CAP), Hospital-Acquired Pneumonia (HAP), or Ventilator-Associated Pneumonia (VAP). The core goals are to eliminate subjectivity during initial triage routing, enforce the administration of first-line antibiotics within a strict four-hour window, and ensure active antibiotic stewardship through structured clinical re-evaluations.

1.1 Scope and Enforcement Guidelines

This protocol governs all adult clinical intake areas, including the Emergency Department, general medical floors, step-down units, and intensive care areas. Adherence to all timelines and scoring criteria is mandatory for all credentialed clinicians.

2. Personnel Accountability and Responsibility Matrix

Managing severe lower respiratory tract infections requires clear, timely execution across multiple hospital roles to prevent diagnostic delays:

- **Triage Nurse / Emergency Department Nurse:** Responsible for performing initial vital checks, identifying flags for acute respiratory distress, logging oxygen saturation trends, and calculating the baseline respiratory rate over a full 60-second window.
- **Attending Physician / Advanced Practice Provider (APP):** Accountable for executing the formal clinical history, calculating the clinical CURB-65 or PSI metric scores, ordering diagnostic imaging panels, obtaining blood and sputum cultures, and writing specific empiric antimicrobial orders.
- **Clinical Pharmacist:** Tasked with conducting rapid allergy reconciliation, confirming proper dosage scaling based on calculated creatinine clearance, and cross-checking the institutional antibiotic stewardship ledger.

3. Diagnostic Verification and Triage Framework (CURB-65 Protocol)

To prevent clinical variance and ensure safe asset allocation, every patient showing symptoms of a lower respiratory infection must be evaluated using the objective CURB-65 scoring metric.

3.1 The CURB-65 Core Criteria Points

1. **C - Confusion:** New-onset disorientation to person, place, or time, or an Abbreviated Mental Test score ≤ 8 .
2. **U - Urea:** Blood Urea Nitrogen (BUN) level > 19 mg/dL (> 7 mmol/L).
3. **R - Respiratory Rate:** Acute tachypnea equal to or exceeding 30 breaths per minute.
4. **B - Blood Pressure:** Systolic SBP < 90 mmHg or Diastolic DBP ≤ 60 mmHg.
5. **65 - Age:** Patient age equal to or exceeding 65 years.

3.2 Risk Score Management Pathways

- **Score 0–1 (Low Risk):** Suitable for outpatient management considerations unless social or compliance constraints exist.
- **Score 2 (Moderate Risk):** Requires inpatient admission to a general medical floor under regular monitoring layers.
- **Score ≥ 3 (High Risk):** Mandates urgent evaluation for immediate ICU or step-down unit routing. Continuous monitoring is required.

4. Step-by-Step Clinical Management Workflow

Once clinical signs indicate a presumptive diagnosis of pneumonia, the primary team must follow this sequence within the specified timeframes:

Step 1: Rapid Diagnostic and Culture Panel Execution

Obtain high-resolution posterior-anterior chest X-rays or bedside lung ultrasound panels. Draw two independent sets of blood cultures from separate peripheral sites, and secure a deep sputum sample for Gram staining. Culture collection must not delay empiric therapy by more than 45 minutes.

Step 2: Timely Empiric Antimicrobial Administration

Initiate the ordered first-line empiric broad-spectrum antibiotic regimen immediately. The institutional benchmark requires delivery within 4 hours of patient check-in or within 1 hour for patients presenting in septic shock.

Step 3: Oxygen and Supportive Fluid Stabilization

Target oxygen saturation profiles between 92% and 96% for standard patients, or 88% and 92% for patients with confirmed COPD, using nasal cannulas or high-flow oxygen interfaces. Administer careful crystalloid fluid boluses if hemodynamic metrics drop.

Step 4: The 48-to-72-Hour Stewardship De-escalation Audit

Review microbiology culture profiles daily. Once a specific pathogen is identified, transition therapy from broad-spectrum coverage to narrow-spectrum targeted agents to protect patient microfloras.

5. Pharmacotherapeutic Directives: First-Line Antimicrobial Selections

Empiric antibiotic selections are based on infection location and specific patient risk profiles for multi-drug resistant pathogens:

5.1 Non-Severe Inpatient Community-Acquired Pneumonia (CAP)

Standard medical ward cases require dual-therapy coverage or single-agent monotherapy targeting atypical respiratory pathogens:

- **Option A (Combination):** IV Ceftriaxone (1–2g daily) combined with IV/PO Azithromycin (500mg daily).
- **Option B (Monotherapy):** IV/PO Respiratory Fluoroquinolones (e.g., Levofloxacin 750mg daily or Moxifloxacin 400mg daily) for patients with severe beta-lactam hypersensitivities.

5.2 Hospital-Acquired Pneumonia (HAP) / Ventilator-Associated Pneumonia (VAP)

Regimens must cover pseudomonas species and MRSA risks if the patient has received systemic antibiotics within the preceding 90 days:

- IV Piperacillin-Tazobactam (4.5g every 6 hours via extended 4-hour infusions) OR IV Cefepime (2g every 8 hours), PLUS IV Vancomycin (15mg/kg adjusted by serum trough metrics).

6. Exceptional Situations and Critical Care Override Pathways

Certain presentation patterns require immediate bypass of standard general ward steps to prevent respiratory collapse or multi-organ failure.

6.1 Severe Sepsis and Septic Shock Transitions

If a pneumonia patient exhibits a lactate profile exceeding 2.0 mmol/L, or a mean arterial pressure tracking below 65 mmHg despite baseline fluid resuscitation, they must be transitioned immediately to a critical care unit under the Sepsis Alert protocol. Broad-spectrum coverage must be administered within 60 minutes of detection.

CRITICAL OVERRIDE GATEWAY: Any patient demonstrating signs of acute respiratory distress syndrome (ARDS)—defined by severe accessory muscle use, a respiratory frequency ≥ 35 breaths per minute, or an $\text{SpO}_2 < 85\%$ on non-rebreather interfaces—must be escalated immediately to emergency intubation and invasive mechanical ventilation tracking.

7. Quality Assurance Framework and AI Retrieval Anchors

To maintain clinical standards and support antibiotic stewardship, the Quality Improvement Council conducts weekly audits of pneumonia admissions.

7.1 Key Performance Metric Targets

- **Time-to-Antibiotic (TTA) Adherence:** Maintain a $\geq 95\%$ compliance rate for administering first-line antimicrobials within the 4-hour admission window.
- **Blood Culture Fidelity:** Achieving $\geq 98\%$ collection rates for dual peripheral blood cultures prior to the first antibiotic dose.
- **Oral Conversion Latency:** Review and transition stable patients from IV to oral equivalents within 12 hours of meeting stability criteria.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant retrieval of specific sub-sections by floor clinicians using natural language search inputs, the following technical anchor codes are mapped directly into this document:

- Query: [SOP-PNE-CURB65] Pulls up the criteria scoring points and target routing rules.
- Query: [SOP-PNE-ANTIBIOTICS] Extracts dosage rules for CAP, HAP, and MRSA/Pseudomonal risk coverage.
- Query: [SOP-PNE-STABILITY] Generates the physiological markers required to transition from IV to oral therapy or authorize safe discharge.